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Spatial Biodiversity Modeling Pipeline and Benchmark

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Contents

1 Introduction **2**

1.1 Motivation 2

1.2 Research Objectives and Scope 3

2 Background **5**

2.1 Biodiversity and how to measure it 5

2.2 Spatial Biodiversity Modeling 7

3 Methods/setup **8**

3.1 Pipeline for Community Level Modeling 8

3.2 Data 8

3.3 Models 8

4 Results/ Case Study **9**

5 Discussion **10**

5.1 Findings 10

5.2 Ethical Discussion 10

6 Future Work/ Applications **11**

7 Conclusion **12**

A Appendix A: Github Repository **12**

B Appendix B: Additional Results **13**

1 Introduction

1.1 Motivation

Biodiversity is defined as "the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."[1] In this sense, biodiversity refers to the biotic component of ecological systems: the variety and variability of living organisms, from plants and animals to fungi, bacteria, and protists. Biodiversity is integral to ecosystem functioning and stability, supporting processes such as nutrient cycling, pollutant breakdown, water regulation, carbon storage, and ecosystem productivity (Stork 2009; Cardinale et al. 2012; Gascon et al. 2015). Furthermore, biodiversity is fundamental to human survival and well-being. It supports food security through pollination, soil fertility, fisheries, and livestock systems (Gascon et al. 2015; Brauman et al. 2020). It also provides biological materials and natural compounds used in medicine and drug discovery, including antibiotics such as penicillin (Gascon et al. 2015). Beyond these material contributions, biodiversity supports regulating ecosystem services such as water purification, climate regulation, and carbon storage, while also contributing to biomass-based energy systems and sustaining human health, livelihoods, and well-being (Brauman et al. 2020; Naeem et al. 2016; Kim et al. 2024).

Despite the overwhelming benefits of biodiversity to humanity and ecosystems, biodiversity is declining at an unprecedented rate. Approximately 74% of the world's terrestrial wilderness has experienced degradation, and approximately 9.6% has been completely lost since the early 1990s (Watson et al. 2016). Marine ecosystems have also been heavily affected, with 66% of the ocean's surface having been significantly impacted by human activities (Díaz et al. 2019). Moreover, it is reasonably likely that most coral reefs will be lost if global temperatures exceed 2°C (Frieler et al. 2012). Furthermore, human activities threaten the survival of approximately 1 million species, with approximately 25% of plants and animals threatened (Díaz et al. 2019). Species extinction rates have also increased and are now estimated to be around 1,000 times higher than the natural background rate (Pimm et al. 2014). Compounding this challenge, conservation efforts are underfunded, with a deficit exceeding half a trillion US dollars annually (Deutz et al. 2020). The acceleration of biodiversity loss and the drastic underfunding of conservation efforts have necessitated a shift towards the prioritization of biodiversity conservation and ecological rehabilitation.

Many initiatives around the world are being adopted to combat biodiversity loss and climate change, which are intricately linked. These initiatives focus on international cooperation, expanding protected areas, reducing pollution, and providing incentives for sustainable practices (IPBES 2019; IPCC 2023). Specific policy initiatives such as the 30x30 target—which aims to protect 30% of the world's land area by 2030—represent critical efforts for ecological recovery. If successful, they could generate large gains and rebounds in biodiversity. Achieving these goals would have significant impacts, including carbon sequestration, improved water quality regulation, and expanded capabilities to mitigate nutrient pollution (Zeng, Koh, and Wilcove 2022). For these initiatives to be successful, it is crucial to have accurate and robust models that can predict biodiversity patterns and trends. These models can inform conservation strategies, identify priority areas for protection, and assess the potential impacts of different conservation actions (Sinclair, White, and Newell 2010).

At their core, biodiversity models are spatial models that relate biodiversity patterns to environmental conditions at particular locations (Ferrier and Guisan 2006). In this thesis, we focus on community-level models (Nieto-Lugilde et al. 2018), which refer to models that predict community-level biodiversity metrics, such as species richness, total abundance or relative abundance, intactness, or compositional similarity (Lamb et al. 2009; Santini et al. 2017). Broadly, community-level modeling approaches can be grouped into three main strategies based on how they produce predictions for community-level biodiversity metrics (Ferrier and Guisan 2006). The first strategy is a bottom-up, or "predict first, assemble later" strategy (Ferrier and Guisan 2006; Pollock et al. 2020). This strategy utilizes species distribution models (SDMs) to predict the spatial distribution of individual species based on a wide range of environmental variables; multiple SDMs are then aggregated or stacked together to estimate community-level biodiversity metrics (Ferrier and Guisan 2006; Distler et al. 2015; Beery et al. 2021). The second strategy

[1] Convention on Biological Diversity (1993) *Article 2. Use of terms.* Available at: https://www.cbd.int/convention/articles/?a=cbd-02 (Accessed: 8 April 2026).

is an "assemble and predict together" strategy (Ferrier and Guisan 2006). This strategy utilizes joint species distribution models (JSDMs) to model multiple species simultaneously, accounting for shared environmental responses and residual species associations (Warton et al. 2015). The final strategy is a top-down, or "assemble first, predict later" strategy (Ferrier and Guisan 2006; Pollock et al. 2020). In this strategy, the model directly predicts the community-level biodiversity metrics rather than first modeling individual species or multi-species assemblage predictions (Pollock et al. 2020).

While each strategy has its own strengths and weaknesses and excels in different contexts, it is unlikely that one modeling strategy will consistently outperform the others (Ferrier and Guisan 2006; Zhang et al. 2019). However, bottom-up approaches are the most extensively researched modeling strategy for community-level biodiversity prediction. Many studies have focused on improving species distribution models, including benchmark evaluations (Norberg et al. 2019; Valavi et al. 2022), reviews of specific models (Waldock et al. 2022), and guidance on best practices (Araújo et al. 2019). Similarly, "assemble and predict together" models have also been the focus of many studies. Joint species distribution models have been developed partly to address the limited treatment of species co-occurrence and biotic interactions in traditional species distribution models (Wilkinson et al. 2019; Escamilla Molgora et al. 2022). Top-down models, on the other hand, are less widely studied and underutilized (Pollock et al. 2020). A significant gap remains in our understanding, comparison, and evaluation of bottom-up and top-down models, both between these modeling approaches and within the different methods they encompass. This highlights the need for further research and methodological development of top-down models (Ferrier and Guisan 2006; Pollock et al. 2020).

Currently, top-down models are often evaluated in individual studies that use different datasets, model types, biodiversity metrics, and evaluation procedures (Mondal and Bhat 2021; Cai et al. 2023; Ngor et al. 2023; Baggström et al. 2025). This can make it difficult to understand the current landscape of top-down models. While there have been some reviews of community-level models as a whole (Nieto-Lugilde et al. 2018), there still remains a lack of best practice guidelines for top-down models, comparable to those available for bottom-up approaches (Araújo et al. 2019). This has prompted the need for a standardized pipeline for models using a top-down strategy to allow for more rigorous testing (Pollock et al. 2020). This pipeline would allow for a more systematic evaluation of different top-down models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a standardized pipeline would enable a more nuanced evaluation of these models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

1.2 Research Objectives and Scope

The primary objective of this master's thesis is to develop a standardized pipeline for top-down models that predict community-level biodiversity metrics and to provide a case study demonstrating the applicability and flexibility of this pipeline. To achieve this objective, a pipeline is first developed that integrates generalized data preprocessing, feature engineering, model training, and evaluation steps for top-down models. This pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches. Next, a case study is conducted using this pipeline to compare the performance of different models on derived biodiversity metrics from the United Kingdom Butterfly Monitoring Scheme (UKBMS) dataset (UK Butterfly Monitoring Scheme 2026), accessed through the Global Biodiversity Information Facility (GBIF)[2]. Models used in this case study will include statistical models, specifically generalized linear models and generalized additive models (Nelder and Wedderburn 1972; Hastie and Tibshirani 1986), as well as machine learning models, including random forests, gradient boosting, and artificial neural networks (Breiman 2001; Friedman 2001; Lek et al. 1996). These results are then analyzed to provide insights into the strengths and weaknesses of different top-down models across different contexts. By applying the pipeline across contexts that vary in spatial scale, environmental predictor sets, and evaluation strategy, this thesis demonstrates its flexibility and general applicability to a range of community-level biodiversity modeling scenarios. This comparative framework also enables a systematic assessment of how different modeling classes (statistical or machine learning) perform under varying analytical conditions, thereby providing insight into their relative strengths, limitations, and appropriate

[2]Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 26 April 2026).

use cases. The key research questions that this master's thesis aims to address are:

1. How do models perform with different strategies of cross-validation, specifically random cross-validation vs. spatial cross-validation?
2. How do models perform in data-limited scenarios?
3. How do different model classes (statistical vs. machine learning) perform compared to each other?

The results are therefore intended to illustrate the flexibility and practical utility of the pipeline, while demonstrating how standardized workflows can support more efficient and reproducible evaluation of top-down biodiversity models.

The remainder of this thesis is structured as follows. Section 2 provides background on biodiversity, including how it is defined and measured in Section 2.1, and reviews the main modeling approaches used to predict biodiversity patterns and trends in Section 2.2. Section 3 describes the methodological framework, including the pipeline design in Section 3.1, the case study dataset in Section 3.2, and the models implemented in Section 3.3. Section 4 presents the results of the case study and illustrates the flexibility of the proposed pipeline. Section 5 discusses the implications of these results for biodiversity modeling and conservation, while Section 5.2 addresses relevant ethical considerations. Section 6 outlines potential future directions and applications for research in this area. Finally, Section 7 concludes the thesis by summarizing the key findings and their implications for biodiversity modeling and conservation.

2 Background

This section introduces biodiversity and how it is measured in Section 2.1 and reviews the community-level modeling approaches used to predict biodiversity patterns and trends in Section 2.2. This background section provides the necessary context for understanding the methods and results presented in this thesis. It also highlights the differences between the modeling approaches as well as the different assumptions and limitations of these models.

2.1 Biodiversity and how to measure it

Biodiversity is a complex and multifaceted concept that encompasses the variety of life on Earth, including the diversity of species, genetic diversity within species, and the diversity of ecosystems (Stork 2009). Humanity has relied on biodiversity for survival for thousands of years as it provides us with food, materials, medicine, water security, and has a substantial impact on human health and well-being (Gascon et al. 2015; Naeem et al. 2016; Brauman et al. 2020; Kim et al. 2024). Large losses of biodiversity would be catastrophic for humanity and ecosystems as it would lead to the loss of ecosystem functions, the collapse of food webs, adversely affect human health, degrade medicine production and discovery, and more (Roe, Seddon, and Elliott 2018). The rate of biodiversity loss is unprecedented, and we are quickly moving towards a sixth mass extinction (Cowie, Bouchet, and Fontaine 2022). To combat biodiversity loss effectively, conservation and ecological rehabilitation efforts require accurate, robust models that can predict biodiversity patterns, identify priority areas, and assess potential conservation outcomes (Sinclair, White, and Newell 2010; Pollock et al. 2020).

To effectively predict biodiversity, it is crucial to have a clear understanding of how biodiversity data is collected and measured. Generally, biodiversity data is collected through field surveys, remote sensing, and citizen science initiatives and other monitoring schemes depending on what is being measured (Proença et al. 2017; Beery et al. 2021; Brun et al. 2024). These approaches or protocols can provide data on species occurrences, abundances, and distributions across different spatial and temporal scales (Proença et al. 2017). Once this raw data is collected, there are three broad categories of biodiversity metrics that are commonly computed: alpha diversity, beta diversity, and gamma diversity. Alpha diversity refers to diversity within a site, such as the diversity within a specific community or ecosystem. Beta diversity refers to the diversity between different communities or ecosystems. Gamma diversity refers to the species diversity across communities or ecosystems within a larger region (Andermann et al. 2022). For the purposes of this thesis, we will focus on alpha diversity, which is the most commonly measured type of biodiversity.

Common alpha diversity metrics include species richness, absolute abundance, relative abundance, Shannon index, arithmetic mean abundance, and geometric mean abundance (Santini et al. 2017; Beery et al. 2021). Species richness is the most commonly used metric and simply counts the number of species in a given area (Ferrier and Guisan 2006). Species richness is a very popular metric because presence-absence data is the most common type of biodiversity data (Özkan Tümer et al. 2026). Absolute abundance measures the total number of individuals of each species at a site, and relative abundance measures the proportion of individuals of each species relative to the total number of individuals in a site (Beery et al. 2021). The Shannon index is another commonly used metric and combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community (Beery et al. 2021). Arithmetic mean abundance is the average abundance of species in a given area, and geometric mean abundance is the exponential of the average of the log abundances of species in a given area (Santini et al. 2017).

For the purposes of this thesis, it is beneficial to express these terms mathematically. Biodiversity metrics are usually calculated at the site level. A site is a spatial sampling unit that is defined according to a sampling protocol. We define a site as r and within a biodiversity dataset there are n sites which make up the region of interest $R = \{r_1, r_2, \ldots, r_n\}$. Across the entire region of interest R there is a set S of unique species. Each site r has a set of species S_r which is a subset of the regional species set S.

The alpha diversity metrics used in this thesis are therefore defined at the site level as follows.

Species richness. Following Beery et al. (Beery et al. 2021), species richness is the number of unique

species in a given site r and can be expressed as:

$$SR_r = |S_r| \,. \tag{1}$$

For reference, regional species richness across R can be expressed as:

$$SR_R = \left| \bigcup_{r \in R} S_r \right| . \tag{2}$$

Absolute abundance. Absolute abundance is the total number of individuals of each species at a given site r and can be expressed as:

$$A_r = \sum_{i \in S_r} a_{i,r} \tag{3}$$

where $a_{i,r}$ is the abundance or count of individuals of species i in site r (Beery et al. 2021). Where $a_{i,r} = 0$ if species i is not present in site r.

Relative abundance. Relative abundance is the proportion of individuals of each species relative to the total number of individuals in a given site r and can be expressed as:

$$p_{i,r} = \frac{a_{i,r}}{A_r} \tag{4}$$

where $a_{i,r}$ is the abundance or count of individuals of species i in site r and A_r is the total abundance of all species in site r (Beery et al. 2021). Where $a_{i,r} = 0$ if species i is not present in site r.

Shannon index. The Shannon index combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community. It can be expressed as:

$$H_r = -\sum_{i \in S_r} p_{i,r} \ln(p_{i,r}) \tag{5}$$

where $p_{i,r}$ is the proportion of individuals of species i relative to the total number of individuals in a given site r which is the relative abundance $p_{i,r}$. The interpretation of the Shannon index is that higher values of H_r indicate higher diversity (Dušek and Popelková 2017).

Arithmetic mean abundance. Arithmetic mean relative abundance is the average abundance of species at a given site r. It can be expressed as:

$$M_r = \frac{1}{SR_r} \sum_{i \in S_r} a_{i,r} \tag{6}$$

where $a_{i,r}$ is the abundance or count of individuals of species i in site r (Santini et al. 2017).

Geometric mean abundance. Geometric mean abundance is the exponential of the average of the log abundances of species in a given area and can be expressed as:

$$G_r = \exp\left(\frac{1}{SR_r} \sum_{i \in S_r} \ln\left(a_{i,r}\right) \right) \tag{7}$$

where $a_{i,r}$ is the abundance or count of individuals of species i in site r (Santini et al. 2017). If $a_{i,r} = 0$ for any species i in site r, then that species is not included in the calculation of the geometric mean abundance for site r.

These metrics each have their own benefits and limitations. Species richness is the most commonly used metric because presence-absence data is the most common type of biodiversity data and it is easy to calculate (Lamb et al. 2009; Özkan Tümer et al. 2026). However, species richness does not account for the abundance of species and can be biased towards rare species (Ferrier and Guisan 2006). Absolute abundance and relative abundance capture more information about the abundance of species in a given area, but they can be dominated by common species (Beery et al. 2021). The Shannon index combines species richness and evenness into a single metric and can provide a more comprehensive measure of biodiversity, but it can be difficult to interpret (Dušek and Popelková 2017). Arithmetic mean abundance and geometric mean abundance are useful for measuring changes in overall populations but can be inflated by very high population density species (Santini et al. 2017). It is important to look at a wide range of metrics when measuring biodiversity as each metric measures different aspects of biodiversity and responds differently to changes in biodiversity (Lamb et al. 2009; Santini et al. 2017).

2.2 Spatial Biodiversity Modeling

3 Methods/setup

3.1 Pipeline for Community Level Modeling

3.2 Data

3.3 Models

4 Results/ Case Study

5 Discussion

5.1 Findings

5.2 Ethical Discussion

6 Future Work/ Applications

7 Conclusion

A Appendix A: Github Repository

Includes the instructions there

B Appendix B: Additional Results

Includes the additional results there

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